THE BINOMIAL MODEL

Learning Outcome

- explain how to value a derivative using a one-period binomial model

The binomial model builds on a simple idea: Over a given period of time, the asset's price will either go up (u) to $S_1^u > S_0$ or go down (d) to $S_1^d < S_0$. We do not need to know the future price in advance, because it is determined by the outcome of a random variable. The movement from S_0 to either S_1^u or S_1^d can be interpreted as the outcome of a *Bernoulli trial.*

Let us denote q as the probability of an upward price movement and $1 - q$ as the probability of a downward price movement. With only two possible outcomes—the price either goes up or down—the sum of probabilities must equal 1. We will also find it useful to define the gross return from an up or a down price move as:

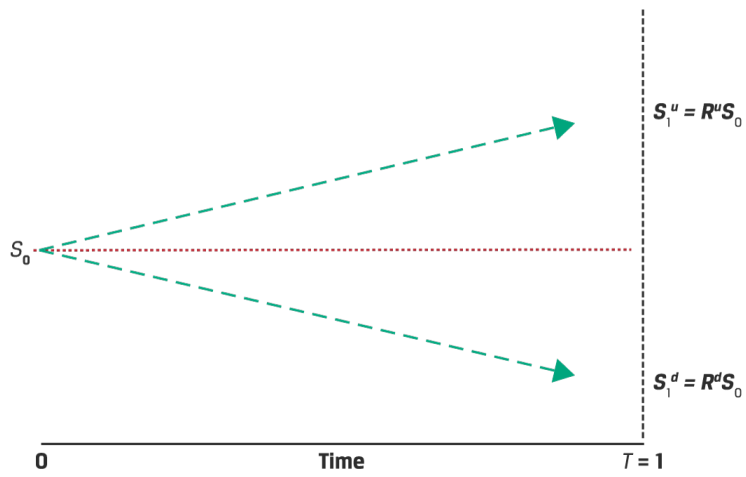
$$R^u = S_1^u/S_0 > 1 \quad (1)$$

$$R^d = S_1^d/S_0 < 1 \quad (2)$$

At first glance, it would appear that knowing q is crucial in determining the value of an option on any underlying asset. However, knowing q is not required; only specifying the values of S_1^u and S_1^d is needed. The difference between S_1^u and S_1^d measures the "spread" of possible future price outcomes. Specifying S_1^u and S_1^d (or R^uS_0 and R^dS_0) determines the *volatility* of the underlying asset, an important factor in valuing options as shown earlier. Simply stated, the size of the up and down price movements should match the underlying asset volatility, as shown in Exhibit 1.

Exhibit 1:

Price Movement for the Underlying Asset



▶ Expand / collapse Extended Description

Binomial models may be extended to multiple periods where the underlying asset price can move up or down in each period. This extension creates a binomial tree, a powerful way to model more realistic price dynamics. A simple one-period binomial model is sufficient to introduce the pricing methodology and the required steps in the option pricing procedure.

Discussion

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Valuing a Derivative Using a One-Period Binomial Model

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